

$ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$
 $\frac{S \vee S'}{\mathbb{Z} \times \mathbb{Z}}$
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 $T: \mathbb{Z} \times \mathbb{Z}$
 $T \hookrightarrow \mathbb{Z} \times \mathbb{Z}$
 $f: \mathbb{Z} \times \mathbb{Z}$
 $f \circ T$

$\pi_1(K)$

$(n \geq 2)$

$\cong \pi_1(M) \oplus \pi_1(N)$

$\{tree\}$

$S^1 \vee S^1$

$Z \neq Z$

n

Example.

$S^2/N = S$

S^r : 0-cell $\geq r$
1-cell $\geq r+1$.

$S^r \cup S^{r+1}$

$\mathbb{R}P^2: \mathbb{Z}_2$
 $\langle a \mid a^2 \rangle$
 $a^2 = 1$
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 $\mathbb{R}P^2: \text{cell } 1, 2, 3, 4$
 $\pi_1(\mathbb{R}P^2) = \mathbb{Z}_2$
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[illegible]
$$\int \frac{f(z)}{g(z)} dz = \sum \frac{2\pi i \cdot \text{Res}(f/g, z_k)}{n_k}$$